

COURSE SYLLABUS

ENGR 1105: Intro to Robotics & Mechatronics

Fall 2026

**RANDOLPH
COLLEGE**

Instructor	Dr. Huan Phan, Assistant Professor of Robotics and Mechatronics Engineering
Email	hphan206@randolphcollege.edu
Class meetings	Mon., Tue., Thu., Fri. • 9:30–10:45 a.m. • Martin Science Building, Room 208
Course dates	October 26–December 14, 2026
Office hours	By appointment; weekly times posted on the course website
Course site	huantianh.github.io/courses/intro-robotics-mechatronics Announcements, readings, lab instructions, assignments, and schedule updates

Course Description

This hands-on course introduces robotics and mechatronics through electronics, Arduino programming, sensors, actuators, MATLAB data analysis, and control. Lectures directly prepare students for weekly Arduino and MATLAB laboratories. Students also complete a separate, self-directed Mechanical Automata project and a final Controller-Based Arduino Project. No previous programming or electronics experience is required.

Course Learning Outcomes

By the end of the course, a successful student will be able to:

1. explain the roles of structures, mechanisms, machines, mechatronic systems, and robots;
2. apply the engineering design process to build, test, and improve a prototype;
3. construct and troubleshoot basic breadboard and Arduino circuits;
4. write introductory Arduino programs using inputs, outputs, timing, conditionals, and sensor data;
5. collect, organize, plot, and interpret sensor data using MATLAB;
6. safely connect sensors, actuators, motor drivers, and power sources; and
7. integrate electrical, programming, sensing, and actuation subsystems into a functioning Arduino-based project and explain its design.

Course Materials

- Course readings, laboratory instructions, and project requirements posted on the course website.
- Arduino Uno-compatible starter kit and a laptop with the Arduino IDE.
- Access to MATLAB for sensor-data analysis and visualization.
- Mechanical Automata and final-project materials approved by the instructor.

- An engineering notebook for sketches, diagrams, code notes, measurements, and test results.
- Safety glasses when instructed.

No textbook purchase is required unless announced by the instructor. Students should not purchase additional project components without instructor approval.

Course Organization and Expectations

The course meets four times per week for 75 minutes. Weeks 1–6 contain lectures and laboratory work. During a normal week, one weekly lecture is divided into Lecture X.1 on Monday and Lecture X.2 on Tuesday. One weekly laboratory is divided into Lab X.1 on Thursday and Lab X.2 on Friday. This structure gives students two class meetings to develop each topic and two class meetings to complete each laboratory. Week 5 is shortened because of Thanksgiving break. Week 7 is reserved for final-project preparation, presentations, demonstrations, and the course project showroom.

- **Project 1 – Mechanical Automata:** A separate, self-directed project completed outside the numbered laboratories. Students independently study a simple mechanism and build a non-motorized prototype that produces repeated animal-inspired motion. The instructor provides requirements, examples, checkpoints, and consultation.
- **Final Project – Controller-Based Arduino Project:** Build any approved Arduino-controlled system with a power source, at least one sensor or user input, at least one output or actuator, and repeatable programmed behavior.

There are six numbered laboratory assignments. Each laboratory may span two class meetings but is submitted and graded as one assignment. There are no separate homework assignments. Students are expected to attend regularly, work safely, contribute to their team, and keep the engineering notebook current.

Assessment and Grading

Assessment category	Weight
Laboratory Activities and Engineering Notebook	30%
Project 1: Mechanical Automata	30%
Final Project: Controller-Based Arduino Project	30%
Peer Evaluation	10%
Total	100%

A	B	C	D	F
90–100%	80–89.9%	70–79.9%	60–69.9%	below 60%

Laboratories and Engineering Notebook

The course includes six laboratory assignments. During a normal week, each laboratory is divided into two sessions, labeled Lab X.1 and Lab X.2, but is submitted and graded as one assignment. Week 5 contains one shortened laboratory session because of Thanksgiving break. Grades reflect preparation, safe participation, completion, troubleshooting, and documentation. Each student must record dated sketches, diagrams, wiring, code notes, measurements, test results, and design changes.

Project 1: Mechanical Automata

The automaton must use no electric motor or electronic controller, produce repeated animal-inspired motion, and include at least one identifiable mechanism. Grading emphasizes function, design, testing, improvement, documentation, and demonstration.

Final Project: Controller-Based Arduino Project

The final project must use an Arduino or compatible controller and include a power source, at least one sensor or user input, at least one output or actuator, and programmed decision logic. The project may be a robot, automated mechanism, interactive device, or another approved mechatronic system. Grading emphasizes safe construction, correct operation, integration, testing, documentation, reliability, and presentation.

Peer Evaluation

Students confidentially evaluate teammates based on preparation, participation, communication, reliability, respect, safety, and contribution. Peer evaluations will inform the grade, but the instructor will review all evaluations and determine the final peer-evaluation grade.

Laboratory Safety and Equipment Care

Follow all laboratory instructions and use tools and components only for their intended purpose. Wear safety glasses when instructed, secure long hair and loose clothing, disconnect power before changing wiring, and never power motors or other high-current devices directly from Arduino input/output pins. Return equipment in organized condition.

Course Policies

Attendance

Regular attendance is expected. In a seven-week course, one absence represents substantial material and project time. Notify the instructor as early as possible and remain responsible for missed work.

Late and Make-Up Work

Unless stated otherwise, late work loses 10 percentage points per calendar day. Contact the instructor before the deadline when illness or an emergency prevents timely completion. Missed laboratory work may require an alternative assignment.

Teamwork and Individual Responsibility

Students may collaborate on team projects, but each student must understand the team's design, wiring, code, and test results. Individual notebook entries and reflections must be the student's own work.

Communication and Conduct

Use Randolph College email and include ENGR 1105 in the subject line. Contribute to a respectful, inclusive, and productive learning environment. Phones and unrelated devices should be put away; laptops may be used for course work.

Randolph College Policies and Resources

Accommodations

Students seeking accommodations should contact Larvail Jones, Coordinator of Access Services, at ljones@randolphcollege.edu or 434-947-8132, then meet privately with the instructor after receiving an accommodation letter.

Title IX and Sexual Misconduct

Faculty members are mandatory reporters. Reports involving sexual misconduct or a campus crime must be shared with the Title IX Coordinator, Jaclyn Beard, at jbeard@randolphcollege.edu or 434-947-8031. The Counseling Center is a confidential resource at counselingcenter@randolphcollege.edu or 434-947-8130.

Honor Code

Students must submit their own work, identify collaborators, cite sources and software, and never invent measurements or test results. Violations will be handled under the Randolph College Honor Code. See <https://www.randolphcollege.edu/studenthandbook/>.

Artificial Intelligence

AI tools may be used for brainstorming, grammar, conceptual explanations, or debugging questions. They may not generate submitted designs, code, diagrams, reports, notebook entries, or reflections unless explicitly authorized. Permitted use must be disclosed, and students remain responsible for accuracy, safety, and originality.

Tentative Course Schedule

Weeks 1–6 contain lectures and laboratory work. During a normal week, one weekly lecture is divided into two lecture meetings and one weekly laboratory is divided into two laboratory meetings. Week 5 is shortened because of Thanksgiving break. Week 7 is reserved for final-project preparation, presentations, demonstrations, and the project showroom. Project 1 is completed separately from the numbered laboratories.

Week	Date	Topic or activity
1	Mon., Oct. 26	Lecture 1.1 – Robotics and Mechatronics Foundations: Sensors, controllers, actuators, and the sense–decide–act model. Project 1 assigned.
1	Tue., Oct. 27	Lecture 1.2 – Robotics and Mechatronics Foundations: Voltage, current, resistance, polarity, power, circuit diagrams, and breadboards.
1	Thu., Oct. 29	Lab 1.1 – Arduino Foundations: Arduino IDE, board connection, digital output, LEDs, and resistors.
1	Fri., Oct. 30	Lab 1.2 – Arduino Foundations: Push buttons, digital input, simple decisions, troubleshooting, and submission. Project 1 concept due.
2	Mon., Nov. 2	Lecture 2.1 – Arduino Programming: <code>setup()</code> , <code>loop()</code> , variables, constants, functions, and readable code.
2	Tue., Nov. 3	Lecture 2.2 – Arduino Programming: Conditionals, loops, timing with <code>millis()</code> , Boolean logic, analog input, and PWM.
2	Thu., Nov. 5	Lab 2.1 – Arduino Programming: Multiple LEDs, button-controlled behavior, timing, and a traffic-light sequence.

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Week	Date	Topic or activity
2	Fri., Nov. 6	Lab 2.2 – Arduino Programming: Potentiometer input, analog measurements, PWM, variable output, testing, and submission.
3	Mon., Nov. 9	Lecture 3.1 – Sensors and Data Collection: Analog and digital signals, range, resolution, calibration, noise, and thresholds.
3	Tue., Nov. 10	Lecture 3.2 – Sensors and Data Collection: Serial Monitor, Serial Plotter, sampling, averaging, mapping, and threshold-based decisions.
3	Thu., Nov. 12	Lab 3.1 – Arduino Sensors: Read and calibrate a light sensor or potentiometer and create an automatic response.
3	Fri., Nov. 13	Lab 3.2 – Arduino Sensors: Use an ultrasonic sensor, collect a dataset for MATLAB, test the system, and submit the lab. Project 1 progress check.
4	Mon., Nov. 16	Lecture 4.1 – MATLAB Sensor Data Analysis: Interface, variables, vectors, scripts, indexing, calculations, and basic plots.
4	Tue., Nov. 17	Lecture 4.2 – MATLAB Sensor Data Analysis: Importing data, plotting, statistics, calibration curves, thresholds, and interpretation.
4	Thu., Nov. 19	Lab 4.1 – MATLAB Sensor Analysis: Import Arduino data, create a script, calculate descriptive values, and produce labeled plots.
4	Fri., Nov. 20	Lab 4.2 – MATLAB Sensor Analysis: Compare trials, create a calibration curve, select a threshold, interpret results, and submit the analysis. Project 1 demonstration or video due.
5	Mon., Nov. 23	Lecture 5.1 – Actuators and Power: DC and servo motors, torque, speed, current, PWM, motor drivers, batteries, grounding, and protection.
5	Tue., Nov. 24	Lab 5.1 – Arduino Actuator Control: Connect and control a servo or DC motor, use sensor input to produce motion, test the system, and submit the shortened lab.
5	Thu., Nov. 26	No class – Thanksgiving break.
5	Fri., Nov. 27	No class – Thanksgiving break.
6	Mon., Nov. 30	Lecture 6.1 – Arduino Project Design: Requirements, block diagrams, subsystem interfaces, power, sensors, outputs, and actuators.
6	Tue., Dec. 1	Lecture 6.2 – Arduino Project Design: Feedback, thresholds, state-based logic, reliability, safety, testing, and troubleshooting.
6	Thu., Dec. 3	Lab 6.1 – Final Arduino Project Integration: Assemble and test the project subsystems and programmed behavior.
6	Fri., Dec. 4	Lab 6.2 – Final Arduino Project Integration: Complete integration, testing, troubleshooting, documentation, and submission.
7	Mon., Dec. 7	Final Project Preparation: Testing, troubleshooting, documentation, displays, and presentation rehearsal.
7	Tue., Dec. 8	Technical Presentations: Teams explain project goals, system design, wiring, code, testing, and results.
7	Thu., Dec. 10	Showroom Setup and Practice: Display preparation, demonstration practice, peer feedback, and final corrections.
7	Fri., Dec. 11	Final Project Showroom: Public demonstrations and project evaluation.
7	Mon., Dec. 14	Course Completion: Final demonstrations if needed, project submission, reflection, peer evaluation, and cleanup.

The instructor may revise this syllabus when necessary. Substantive changes will be announced in class and posted on the course website.